PROJECT 2: CONFIDENCE INTERVALS

Shanteria Johnson

MATH 2050 – STATISTICAL DATA ANALYSIS

NEGAR ORANGI-FARD

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Part 1: Data Collection

I collected home sale price data from Zillow.com, focusing on listings in Gwinnett County, Georgia. I selected townhouses for this study. To keep the data consistent, I filtered the search to include homes with 4 bedrooms and at least 3 bathrooms that are currently available. From Zillow, I searched for recently available townhomes in Gwinnett County and selected 25 homes at random from the search results. I made sure each selected home had a sale price and was not a rental listing.

Table of Townhomes Sales Prices (USD):

Townhomes	Sale Price (USD)
1	$525,000
2	$474,900
3	$399,000
4	$374,999
5	$699,000
6	$889,900
7	$459,905
8	$895,000
9	$435,045
10	$499,000
11	$418,605
12	$385,000
13	$456,045
14	$415,305
15	$433,055
16	$859,000
17	$472,930
18	$472,780
19	$493,730
20	$470,580
21	$761,000
22	$491,530
23	$473,530
24	$471,630
25	$809,000

Part 2: Describe the Data

1. **Define the Random Variable X:**

   X represents the sale price (in dollars) of a randomly selected townhomes in Gwinnett County. It is a numerical variable.

2. **Estimated Distribution for the Sample Mean:**

   Since there are 25 random selected townhomes, the Central Limit Theorem will be used. The sample mean is approximately normal distributed.

3. **Descriptive Statistics (Calculated in R):**
   Using RStudio, I entered the 25 prices and calculated the following:

```
# Step 1: Enter home sale prices into a vector

prices <- c(525000, 474900, 399000, 374999, 699000, 889900, 459905, 895000,

        435045, 499000, 418605, 385000, 456045, 415305, 433305, 859000,

        472930, 472780, 493730, 470580, 761000, 491530, 473530, 471630,

        809000)

# Step 2: Calculate mean and standard deviation

mean_price <- mean(prices)

std_dev <- sd(prices)

# Print the results

mean_price

std_dev
```

   Results:

```
> mean_price

[1] 541428.8

> std_dev

[1] 166769.7
```

Part 3: Find the Confidence Interval

1. **Error Bound for the Mean (EBM):**

To calculate the Error Bound for the Mean, we use the formula:

EBM = $z^* \times (s / \sqrt{n})$

Where:

1. z^* = critical value (based on confidence level)
2. s = sample standard deviation
3. n = sample size

EBM = $1.645 \times (166{,}769.67 / \sqrt{25}) \approx 54{,}862.34$

2. **90% Confidence Interval**

Using the sample mean $\bar{x} = 541{,}428.76$ and EBM = 54,862.34:

$\bar{x} \pm \text{EBM} = 541{,}428.76 \pm 54{,}862.34$

Confidence Interval (CI) = (486,566.42, 596,291.10)

R Code Used:

```
prices <- c(525000, 474900, 399000, 374999, 699000, 889900, 459905, 895000,
        435045, 499000, 418605, 385000, 456045, 415305, 433305, 859000,
        472930, 472780, 493730, 470580, 761000, 491530, 473530, 471630,
        809000)

mean_price <- mean(prices)
std_dev <- sd(prices)
n <- length(prices)
z_star <- qnorm(0.95)
```

```
ebm <- z_star * std_dev / sqrt(n)
lower_bound <- mean_price - ebm
upper_bound <- mean_price + ebm

c(lower_bound, upper_bound)
```

3. Graph (Shaded Confidence Interval Curve):

```
# Sample statistics
mean_price <- mean(prices)
std_error <- sd(prices) / sqrt(length(prices))

# Confidence interval (90%)
z_star <- qnorm(0.95)
lower_bound <- mean_price - z_star * std_error
upper_bound <- mean_price + z_star * std_error

# Create x-values for normal curve
x_vals <- seq(mean_price - 4*std_error, mean_price + 4*std_error, length.out = 1000)
y_vals <- dnorm(x_vals, mean = mean_price, sd = std_error)

# Plot the normal curve
plot(x_vals, y_vals, type = "l", lwd = 2, col = "black",
    main = "90% Confidence Interval for Mean Home Price",
    xlab = "Home Sale Price", ylab = "Density")

# Add shaded area for 90% CI
```

```
x_shade <- seq(lower_bound, upper_bound, length.out = 100)

y_shade <- dnorm(x_shade, mean = mean_price, sd = std_error)

polygon(c(lower_bound, x_shade, upper_bound),

    c(0, y_shade, 0),

    col = rgb(0, 0, 1, 0.2), border = NA)


# Add vertical lines for the bounds

abline(v = c(lower_bound, upper_bound), col = "blue", lty = 2)
```

Question #11:

Some students think that a 90% confidence interval contains 90% of the data. Use the list of data on the first page and count how many of the data values lie within the confidence interval. What percent is this? Is this percent close to 90%? Explain why this percent should or should not be close to 90%.

A 90% confidence interval does not mean that 90% of the individual data points are inside it. Instead, it means that if we were to take many samples and compute a confidence interval from each one, about 90% of those intervals would contain the true mean of the population. It's about the accuracy of the interval estimate for the population mean, not how much data falls inside the range.

Question #12:

In two to three complete sentences, explain what a confidence interval means (in general), as if you were talking to someone who has not taken statistics.

A confidence interval gives us a range of values where we believe the true average (mean) might be. For example, if we use a 90% confidence interval, it means we are 90% sure that the actual average value falls somewhere within that range.

Question #13:

In one to two complete sentences, explain what this confidence interval means for this study.

For this study, the 90% confidence interval tells us that we are 90% confident the average sale price of all similar townhomes in Gwinnett County falls between $486,566.42 and $596,291.10 based on the sample of 25 homes.

Part 4: Use the Data to Construct Confidence Intervals

1. Using the sample mean of $541,428.76, sample standard deviation of $166,769.67 and sample size of 25, the following confidence intervals were calculated using RStudio:

Confidence Level	Error Bound (EBM)	Confidence Interval (USD)
80%	$42,744.79	(498,683.97, 584,173.55)
95%	$65,372.51	(476,056.25, 606,801.27)
99%	$85,914.04	(455,514.72, 627,342.80)

Question #15:

What happens to the EBM as the confidence level increases? Explain why this happens.

As the confidence level increases, the Error Bound for the Mean (EBM) also increases. This happens because higher confidence means we want to be more certain that the intervals include the true mean, so we have to allow for wider range.

Question #16:

Does the width of the confidence interval increase or decrease? Explain why this happens.

The width of the confidence interval increases as the confidence level increases. This happens because to be more confident that the interval includes the true mean, we need to capture more possible values, which makes the interval wider. A higher confidence level uses a larger z* value, stretching the range on both sides of the sample mean.